

The background is a dense collage of hexagonal tiles, each featuring a different RStudio ecosystem logo. Visible logos include: lubridate (clock and calendar), forcats (cats), tidyr (sweeping brooms), tidyverse (dots), tidymodels (curved lines), TUNE (green text), tidyposterior (bell curves), readr (document icon), rmarkdown (quill pen), R Studio (blue circle with R), broom (broom icon), WORKFLOWS (circular flow diagram), flexdash (network diagram), rvest (scissors), ggplot2 (network diagram), parsnip (leaf), recipes (cupcake), Shiny (cursive text), modeldb (bar chart), and various others like lubridate, forcats, tidyr, tidyverse, tidymodels, TUNE, tidyposterior, readr, rmarkdown, R Studio, broom, WORKFLOWS, flexdash, rvest, ggplot2, parsnip, recipes, Shiny, and modeldb. The tiles are arranged in a grid-like pattern, creating a colorful and busy background.

Getting Started with Codex

MCB Consulting Group (MKTG 411)

Today's Learning Goals

By the end of class, you should be able to:

- Set up Codex and understand your access options.
- Improve prompts to answer a marketing question with R summaries and charts.
- Use explanations and feedback to build your understanding.
- Check campaign comparisons and AI conclusions, and try a plugin.
- Use the Lecture Setup Skill to organize materials from Canvas.

Simple Tasks

Codex is an AI agent that can work with files and programs on your computer.

When you need help with...	A useful request
Understanding Code	"Explain what each line of this code does."
Fixing an Error	"Explain this error and suggest the simplest fix."
Choosing a Function	"Which tidyverse function should I use to keep only Email orders?"
Making a Plot	"Help me make a bar chart of revenue by marketing channel."

You supply the prompt and evaluate the result.

Complex Tasks

When you need help with...	A useful request
Brainstorming	"My client wants to identify the most valuable customers. I think we should start by identifying who made the most recent purchases. Can you identify any shortcomings with this approach, and suggest alternative definitions of 'valuable' customers? Then we'll identify the pros and cons of each definition together."
Vibe Coding	"Based on this analysis of most valuable customers, I want to build a dashboard that allows decision makers to easily query the data and visualize customer characteristics. Let's start by outlining an implementation plan and then executing on that plan. I'll rely on you to write the back end code, so please provide me with a way to access the front-end dashboard to evaluate it and provide feedback."

Installation and Sign-In

1. Open the official [desktop setup guide](#). Download and install the version for **Windows or macOS**.
2. Open the desktop app and sign in with your **ChatGPT account**.
3. Select **Codex** from the product menu.
4. Send a first message: “I’m learning R for marketing analytics.”

Codex is part of the ChatGPT desktop app. Keep **R** and **RStudio** installed for running and checking your course code.

Setup checked September 7, 2026. [Windows setup help](#).

Our Plan for Access This Semester

- Use your BYU student email to setup an account and/or login
- Claim the student credits on the next slide
- CES ChatGPT Enterprise accounts should give a credit allocation
- Check your usage and request credit limit increases!
- You can also use occasional free resets

Small, focused requests use less capacity than long tasks with many files.

[Official plans and usage guidance](#). Checked September 7, 2026.

Your Student \$100 in Codex Credits

1. Open [Codex for Students](#) and select Claim now.
2. Sign in with the ChatGPT account you use for class. Use your university email to verify your student status.
3. After approval, check your usage page for 2,500 credits (\$100).

(I'm hoping this works even if you are on an enterprise account.)

[Student offer and eligibility](#). Checked September 7, 2026.

Your First Project

- 1. Download `hillstrom-practice.zip` from the Codex lecture on Canvas.
- 2. Create `MKTG-411/codex-practice` and extract the ZIP's contents directly into it: `hillstrom.csv`, `demo.qmd`, and `hillstrom-data-notes.md` together.
- 3. Open `demo.qmd` in RStudio. Set the working directory to `codex-practice`.
- 4. Open that same folder as a Codex project. Start a new task: "Describe the contents of `demo.qmd`".

Part of the app	What you do there
Project and task list	Navigate your folders and conversations.
Message box	Name a file, describe a goal, and give follow-up instructions.
File or change view	Open the saved output and inspect edits.

Saving Files Before Codex Makes Changes

- SAVE YOUR WORK **BEFORE** PROMPTING CODEX.
- Codex works with the saved file on your hard drive.
- If you have unsaved edits, they exist only in RStudio.
- If Codex makes changes to an unsaved file, you can lose your unsaved edits.
- Saving your copy after Codex makes changes can overwrite Codex's changes.
- **Let's see an example of what NOT to do...**

Model Settings for the Demo

Before sending your first prompt, select:

- **Model: Luna**
- **Reasoning effort: Medium**

Use these settings throughout today's demonstration to keep credit usage low.

The Hillstrom Campaign Decision

A retailer wants to generate sales using an email campaign.

Given the data, which email, if either, should it repeat? What evidence supports that choice?

The data contain 64,000 customers randomly assigned to:

- An email featuring men's merchandise.
- An email featuring women's merchandise.
- No email.

We observe purchases and spending over the following **two weeks**.

Discuss: How would we determine whether a campaign should be repeated?

Our First Prompt

Add code to make a bar chart comparing average spend among customers who purchased in the two email campaigns.

Run the prompt and inspect the saved code and chart.

- Which campaign looks better by this measure?
- What does this plot not account for?
- Does this answer the retailer's business question?

Add A Clearer Goal and Relevant Context

Prompting tip: Explain the decision and ask for clarification.

Our goal is to establish whether an email campaign increased revenue. Does our first chart answer that question? Explain any mismatch and ask us one question about how to measure success before changing the analysis.

Does AI identify any key issues?

[OpenAI: clear instructions and relevant context.](#)

A More Useful Deliverable

Prompting tip: Refine the goal, specify the desired output and define an example of its structure.

(Adapt the following as needed):

Revise demo.qmd to answer our business question using numerical summaries and a chart. Include all email campaign groups (including no email). The numerical summaries should be contained in a table with one row per group. Include customer counts, purchase rate, and revenue per targeted customer. Keep the raw data unchanged. Use readable tidyverse code.

Example table headings:

Group / Customers / Purchase rate / Revenue per customer

Specific Feedback on the Output

Prompting tip: Name the problem you actually observe.

Compare the revised output with your first chart.

- Are the units and the two-week period clear?
- Can a reader identify the comparison group?
- Does the written recommendation follow from the numbers?

We need to make the following improvements.... Make those changes and explain back to me how the change helps answer our business question. Make sure to preserve the portions of the analysis that are already correct.

Class contribution: Agree on one specific piece of feedback before sending it.

Honest Conclusions

Prompting tip: Supply evidence and invite disagreement.

The analysis so far suggests that the Men's merchandise email raised revenue per customer significantly compared to the no email group. We could recommend that campaign to the client, but does the evidence support that that campaign truly increases revenue? Give a critical evaluation, and state any additional evidence we need to give a strong recommendation.

[Huang et al. \(2024\): limits of self-correction without external feedback.](#)

Revision and Simplification

Prompting tip: Ask for a focused revision.

Review demo.qmd and remove non-essential complexity. Keep the summaries, chart, and checks needed for our business question. Use readable tidyverse code. Explain each change and rerun the R code chunks to verify that the numerical results stay the same.

Review together:

- Which change makes the analysis easier to understand?
- Did we accidentally lose an important check or limitation?
- If the work was already simple, was any change necessary?

[Madaan et al., Self-Refine \(2023\)](#). The simplification phrase is course guidance, not a guarantee of improvement.

Our Recommendation to the Retailer

Write your own answer first, then ask Codex for a concise revision.

I've written a short recommendation in the .qmd file. Revise it to be more professional and ensure that I am not making claims that the data don't support. Use only results we have checked together.

Compare that answer with your initial recommendation.

Discuss: Which improvement to the prompt changed the business answer most? Which change only improved how we communicated it?

Learning with Codex

Choose the kind of help you need:

“Give me one hint about why including nonbuyers changes our campaign comparison.”

“Explain what `|>` does in our Hillstrom code. Use one calculation to illustrate it.”

“Quiz me on inner joins vs. left joins, one question at a time. Once I consistently answer your questions correctly, give me three hypothetical join scenarios, where I need to determine which type of join to use.”

“Here is my explanation of the code. What did I misunderstand?”

Checking AI Work

Ask AI to track the exact lines it changes and report back on those changes!

- **Review:** Read the changed lines and ask what each change accomplishes if it isn't apparent.
- **Run:** Run the R code chunks in the saved document in RStudio. Check errors and output.
- **Verify:** Do calculations by hand, or ask for a different method to verify the result.
- **Interpret:** Does our campaign comparison answer the business question?

Remember: You are responsible for understanding the work you submit.

Plugins, Skills, and Connectors

Component	What it does	Two examples
Plugin	An installable package that adds capabilities, such as skills or connections to tools.	Spreadsheets: work with Excel files. Chrome: work in your browser.
Skill	Reusable instructions and resources for a particular task or workflow.	Lecture Setup: organize Canvas materials. OpenAI Docs: consult official product documentation.
Connector	A connection that lets Codex read information or take authorized actions in an external service.	Google Drive: access shared documents. GitHub: access code repositories.

A plugin can include skills, connectors, or both.

[Official plugin, skill, and connector definitions](#). Checked September 9, 2026.

Try Another Prompt

“I’m having trouble differentiating between plugins, skills, and connectors in Codex. Give me a simple explanation of the similarities and differences between each one, and include an analogy.”

Spreadsheets Plugin: Revenue Check

1. Open **Plugins**, search for **Spreadsheets**, and open its listing.
2. Install it if needed. Review any connection or permission requests.
3. Start a new task in your practice project and name the plugin:

“Use the Spreadsheets plugin to create **revenue-check.xlsx** with Email orders of \$120 and \$80. Include a SUM formula for the total.”

Open the result and check the formula. If the plugin is unavailable on your account, follow the instructor’s demonstration.

[Plugin installation and use](#). Availability can vary by account.

Chrome Access to Canvas

1. In the desktop app, open **Settings > Computer Use** and select **Chrome**.
Follow the prompts to install the browser plugin and ChatGPT extension.
2. Confirm Chrome shows **Manage**. In that Chrome profile, sign in to Canvas yourself and open our course.
3. Start a Codex task, choose **@Chrome**, and ask:
| “Read my open MKTG 411 Canvas page and tell me which lecture modules you see.”

When prompted, choose **Allow once** or **Allow for this site** for Canvas. Keep passwords and verification codes in the browser sign-in flow.

[Browser extension setup and site permissions](#). Availability depends on your account and setup.

A Reusable Skill: Lecture Setup

The browser plugin provides access. A **skill** supplies a repeatable workflow.

1. Download [lecture-setup.zip](#) with this lecture's materials and extract the `lecture-setup` folder into your course folder.
2. Open that local folder in Codex and ask:

“Install the `lecture-setup` folder as a personal Codex skill.

The skill identifies the lecture, downloads student materials, and places them in the corresponding local folder while preserving your existing work.

Lecture Setup in Action

Open your local **MKTG-411** folder as a Codex project. Select **@Chrome** and login to a Canvas course tab, then send:

“\$lecture-setup: Set up Lecture 7, the Live Workshop in the Data Unit, from the MKTG 411 Canvas course. Use the current MKTG-411 project folder as the course root. Download the lecture materials into a lecture folder with an appropriate name.”

- Supply the lecture number/title you need and answer any course-match question.
- Check the saved files and `lecture-materials.md` for missing or online-only items.
- Next time, name a different lecture and reuse the same skill.

Five-Minute Practice

The retailer now asks whether **website visits** tell the same story as revenue.

Using the same Hillstrom data, work with a partner to request a compact comparison of all three groups and a brief answer. Keep demo.qmd's revenue analysis available for comparison.

- Decide what a fair comparison of visits would show.
- Write a prompt that names the goal, context, and desired output.
- Check one result independently, then give Codex specific feedback.
- Explain whether your finding changes the recommendation and why.

Be ready to share one prompt improvement and one check you performed.